

PROJECT APPRAISAL

Module 3

Discounting and Alternative Investment Criteria



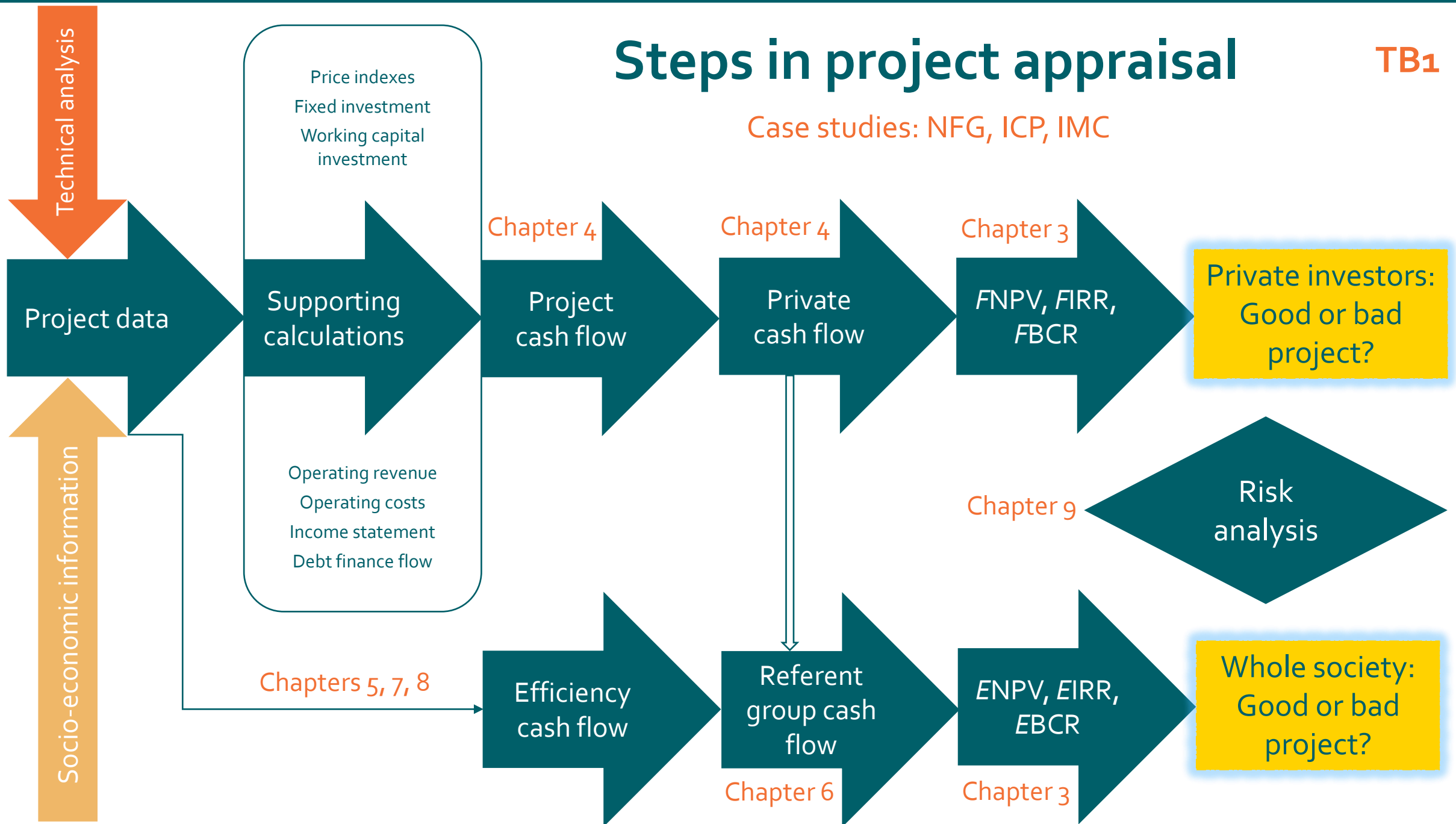
Outline – Module 3

- Future values and the myth of compound interest
- Discounting and financial discount rates
- Total present value and typical financial functions
- Alternative investment criteria
 - Net present value (NPV)
 - Internal rates of return (IRR, MIRR)
 - Benefit to cost ratio (BCR)
 - Payback period (PP)
- Financial sustainability of a project
- An illustrative example

Steps in project appraisal

TB1

Case studies: NFG, ICP, IMC



Future values

The power of compound interest



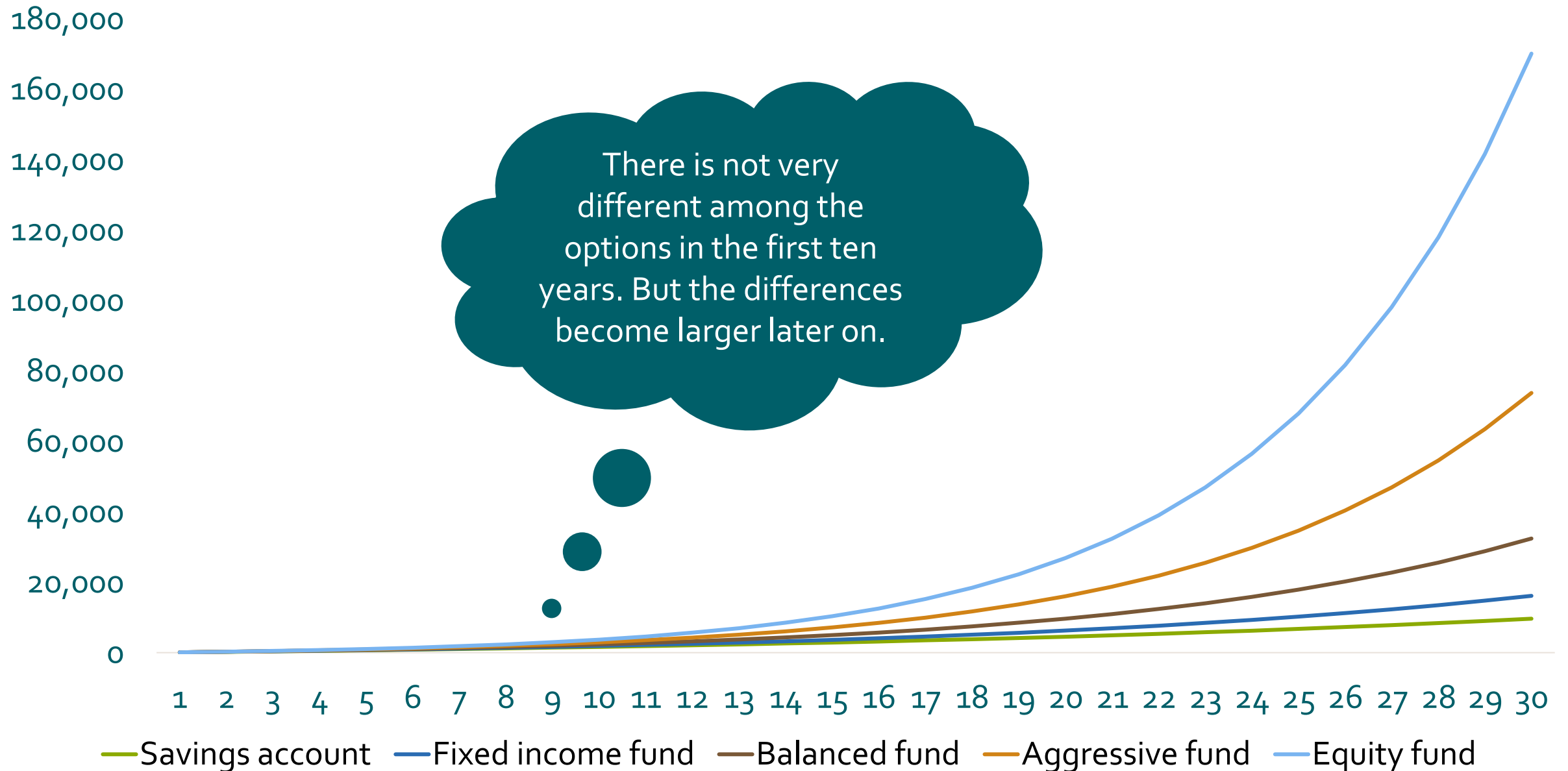
Why should you save money as soon as possible?

Suppose you'll earn an income of 20 million per month and plan to save a half every month in the next 30 years. You might choose one of the following options. Will you decide to keep your money in a savings account?

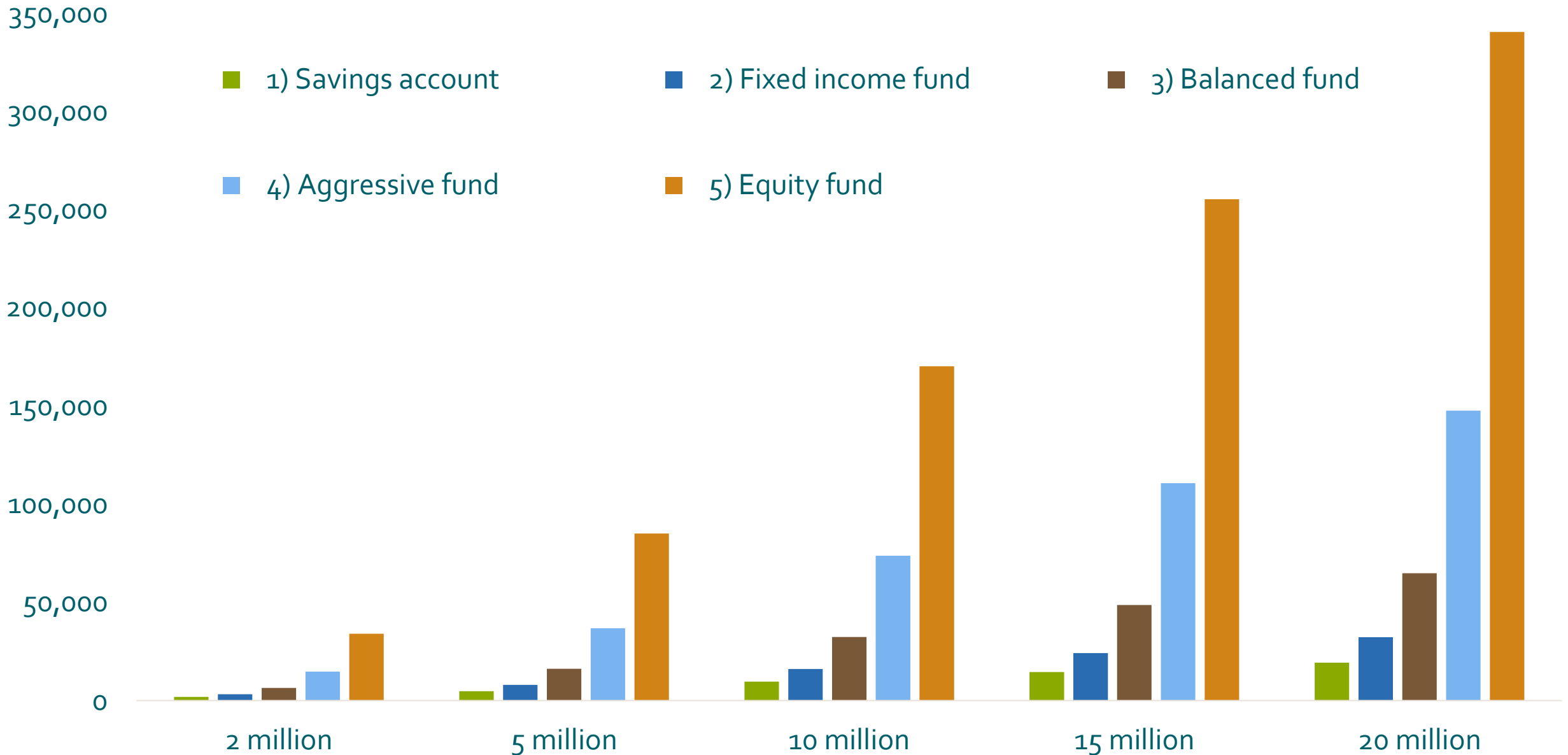
Investment	10	million per month	REMARKS: In the early years we do not see any significant difference. After about 10 years, the differences become larger and larger (Figure 1). If we increase the monthly investment amount, the differences are remarkably huge (Figure 2).			
	120	million per year				
Inflation rate	5%	per year				
Interest rates	Real	Nominal	Option	Asset value after 30 years	Comparison	
	0.8%	5.8%	per year	1) Savings account	9691	1.0
	3.3%	8.5%	per year	2) Fixed income fund	16173	1.7
	6.7%	12.0%	per year	3) Balanced fund	32435	3.3
	10.5%	16.0%	per year	4) Aggressive fund	73819	4.6
	14.3%	20.0%	per year	5) Equity fund	170191	17.6

Future values of 10 million investment per month					=FV(interest rate,t,-120)			
Year	1	2	3	4	5	10	20	30
Savings account	127	261	403	554	713	1658	4571	9691
Fixed income fund	130	271	425	591	771	1932	6299	16173
Balanced fund	134	285	454	642	854	2359	9684	32435
Aggressive fund	139	301	488	705	957	2968	16061	73819
Equity fund	144	317	524	773	1072	3738	26883	170191

Future values over time of 10 million investment per month



ASSET VALUES AFTER 30 YEARS (MILLION)



Discounting

What is the correct discount rate?



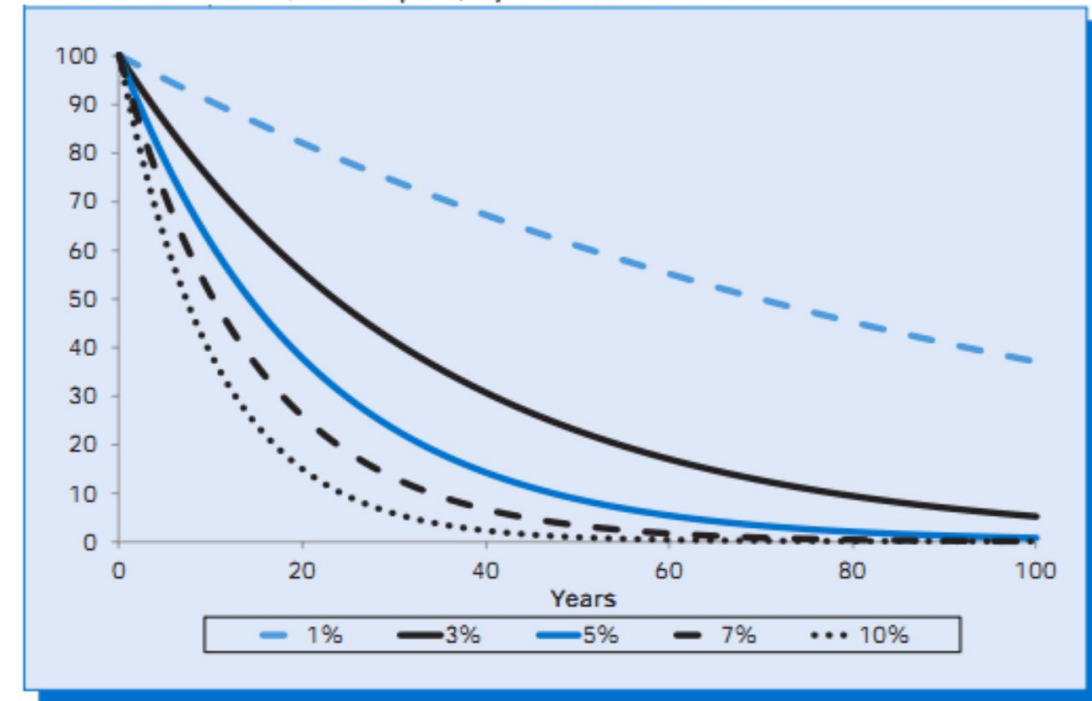
What is discounting?

- Most projects are characterized by large initial investment outlays, but larger benefit flows often occur later in the future.
- A cost of \$100 now is not equivalent to a cost of \$100 in 10 or 20 years from now, due to:
 - Inflation
- But even if the \$100 value is expressed in real (**inflation-adjusted**) terms, the value of \$100 in the future will generally not be the same as its value today.
 - Opportunity cost of money
 - Uncertainty about the future
 - Impatience, the natural human tendency to focus on the present more than the future.
- So, beyond adjusting for inflation, most economists believe that a further adjustment is necessary to compare present and future impacts.
- This adjustment is known as **discounting**, that essentially “devalues” any impacts that occur in the future as less relevant than a similar impact that occurs now. The formula to calculate the present, or discounted, value of a future benefit or cost: $PV(X_t) = X_t / (1 + r)^t$, where r is the discount rate.

What is discounting?

- The higher discount rates dramatically reduce the relevance of impacts that occur just a few decades in the future.
- With a 5 percent discount rate it would not be worth spending even \$1 now to avoid a damage of \$100 that occurs 100 years in the future.
- With a 10 percent discount rate it would not be worth spending even \$1 now to receive a benefit of \$100 that occurs 50 years in the future.
- The choice of a discount rate is an important decision in any CBA or project appraisal.

Present Value of a \$100 Impact, by Discount Rate



Present Value of a \$100 Impact, by Discount Rate

Years into the Future	Discount Rate				
	1%	3%	5%	7%	10%
0	\$100.00	\$100.00	\$100.00	\$100.00	\$100.00
10	\$90.53	\$74.41	\$61.39	\$50.83	\$38.55
20	\$81.95	\$55.37	\$37.69	\$25.84	\$14.86
30	\$74.19	\$41.20	\$23.14	\$13.14	\$5.73
50	\$60.80	\$22.81	\$8.72	\$3.39	\$0.85
100	\$36.97	\$5.20	\$0.76	\$0.12	\$0.01

A high discount rate will highly favor the present over the future, while a low discount rate will give more weight to future costs or benefits.

Discount rates

- The discount rate is a key variable in applying investment criteria for project selection. Its correct choice is critical given the fact that a small variation in its value may significantly alter the results of the analysis and affect the final choice of a project.
- The discount rate is the **opportunity cost of funds** that are invested in the project. It depends upon the viewpoints of analysis.
- For instance, when a project is being appraised from the point of view of the equity holders, the relevant cost of funds is the **return to equity** that is being earned in its best alternative use.
- When we conduct the efficiency analysis of a project, the relevant discount rate is the **economic opportunity cost** for the country.

Discount rates

Y	Year	0	1	2	3
	Cost	-1000			
	Benefit		445.26	445.26	445.26
	NB _Y	-1000	445.26	445.26	445.26

An equity owner will certainly earn an annual rate of return 16% in three years from an initial investment of \$1000 (the best alternative option): **PMT**.

X	Year	0	1	2	3
	Cost	-1000			
	Benefit		400	500	600
	NB _X	-1000	400	500	600

Suppose the equity owner is considering to invest this amount in a **project** and will earn \$400, \$500, and \$600, respectively. How will he make decision?

There are three ways to make decision:

- 1) Check whether the project will earn an annual rate of return higher than 16%?
- 2) Check whether the net present value of the project will be larger than that of the best alternative option at 16%? (= 0)
- 3) Use increamental cash flow (i.e., $NB_X - NB_Y$), then calculate the internal rate of return?

Discount rates

TIPV	Year	0	1	2	3
	Cost	-1000			
	Benefit		400	500	600
	NB _{TIPV}	-1000	400	500	600

EPV	Year	0	1	2	3
	Cost	-400			
	Benefit		154.47	254.47	354.47
	NB _{EPV}	-400	154.47	254.47	354.47

A project invests \$1000, in which the equity owner contributes \$400 with the opportunity cost of **16%** per annum and borrows the rest from a bank with an interest rate of **11%** per annum. What is the discount rate in this case?

The equity owner will not undertake the project without the loan of \$600 from the bank. The loan \$600 is a receipt and the annual debt repayments \$245.53 are expenditures. What is the discount rate in this case?

We need to calculate the opportunity cost of “capital” in each case:

- 1) In case 1, the opportunity cost is $0.6 \times 11\% + 0.4 \times 16\% = 13\%$ per annum (TIPV).
- 2) In case 2, the opportunity cost is **16%** per annum (EPV).

Discount rates

- In **efficiency analysis of any project**, the social discount rate is the economic opportunity cost of capital for the country. It depends on the sources of funds.
 - First, funds that would have been invested in other investment activities have now been displayed by the project. The cost of these funds is the **gross of tax return** that would have been earned by the best alternative investments, which are now foregone.
 - Second, funds come from different categories of savers in the economy who postpone some of their consumption in the expectation of getting a return on their savings. The cost of these funds is the **cost of postponing this consumption**.
 - Third, some funds may be coming from abroad, that is from foreign savers. The cost of these funds is the **marginal cost of foreign borrowing**.
- If the project use all of these funds, the social discount rate will simply be a **weighted average of the costs of funds** from these alternative sources.

Discount rates

In **efficiency analysis of public projects**, there are two different main approaches for determining how a social discount rate should be chosen.

- 1) One approach to determine the social discount rate is to set it equal to the **rate of return on low-risk investments** such as government bonds. The rationale behind this is that funds used for a public project could otherwise be invested with interest to provide society with greater resources in the future. However, the rates of return on government bonds vary over time.
- 2) A different approach to determining the social discount rate is based on two justifications for discounting:
 - a) The natural human tendency to prefer the present over the future. This is known as the **pure rate of time preference** (δ).
 - b) Assuming that real economic growth continues, people in the future will be richer than people now. Therefore, utility is assumed to be a diminishing marginal function of consumption.

Discount rates

These two factors can be combined to estimate the social discount rate, or **social rate of time preference (S RTP)**, as:

$$\text{S RTP} = \delta + \varepsilon * c$$

where δ is the pure rate of time preference, c is the annual growth rate of consumption, and ε is the elasticity of the marginal utility of consumption.

Empirical studies use a value of $c = 1.3$ percent, and values of ε of about 1 to 2, while the value of 1.4 is the most popular.

In 2001, over 2000 economists were surveyed for their opinion on “*the appropriate real discount rate to be used for evaluating environmental projects over a long time horizon*”. The average response was 4 percent, with a median value of 3 percent.

Present value

Different flows of net benefit in the future



The present value of any project's cash flow

A project X that invests NCF_0 at year 0, and generates annual net benefits (from year 1): $NCF_1, NCF_2, \dots, NCF_n$ will have a total present value as shown in the below formula (where $DF_t = 1/(1+r)^t$ is discount factors, i.e. a present value of \$1 in year t with a constant discount rate r):

$$PV(X) = \sum_{t=1}^n \frac{NCF_t}{(1+r)^t} \xrightarrow{\text{Excel}} =NPV(r, NCF_1:NCF_n)$$

As the discount rate, r , increases, the total present value of the project X will be lower.

If we compare the total present value with the initial investment NCF_0 , we will define the net present value: $NPV = PV(X) - NCF_0$:
 $=NPV(r:NCF_1:NCF_n) + NCF_0$.

	A	B	C	D	E	F	G	H
4		Any cash flow, using the NPV function!						
5		Year	0	1	2	3	4	5
6		1. Receipts		0	3247	4573	3525	2338
7		2. Expenditures		2120	1000	1000	1000	1000
8		3. Net cash flows		-2120	2247	3573	2525	1338
9		4. Discount factors @6%		0.94	0.89	0.84	0.79	0.75
10		5. Present values [=4*3]		-2000	2000	3000	2000	1000
11		6. Total present value	6000					
12		NPV function in Excel:	=NPV(6%,D8:H8)					
13			NPV(rate, value1, [value2], [value3], ...)					

	A	B	C	D	E	F	G	H
14		Year	0	1	2	3	4	5
15		1. Receipts		0	3247	4573	3525	2338
16		2. Expenditures	5000	2120	1000	1000	1000	1000
17		3. Net cash flows	-5000	-2120	2247	3573	2525	1338
18		4. Discount factors @6%	1.00	0.94	0.89	0.84	0.79	0.75
19		5. Present values [=4*3]	-5000	-2000	2000	3000	2000	1000
20		6. Net present value [NPV]	1000					
21		NPV function in Excel: =NPV(6%,D17:H17)+C17						
22		NPV(rate, value1, [value2], [value3], ...)						

The present value of a perpetuity

- ◆ What is a perpetuity?
 - Denote a perpetuity as **P**
 - A perpetuity is the equal annual net benefit that goes on for ever.
 - Suppose we have the following perpetuity:

Year	0	1	2	3	...	∞
NCF		P	P	P	...	P

- With an annual constant discount rate r , total present value of a perpetuity is defined as:

$$PV(P) = \frac{P}{r}$$

The present value of a perpetuity

◆ Proof:
$$PV(P) = \frac{P}{(1+r)} + \frac{P}{(1+r)^2} + \dots + \frac{P}{(1+r)^\infty}$$

Denote $a = \frac{P}{(1+r)}$ and $X = \frac{1}{(1+r)}$

$$PV(P) = a + aX + aX^2 + aX^3 + \dots + aX^\infty \quad (1)$$

If we multiply both sides of equation (1) by X , we get:

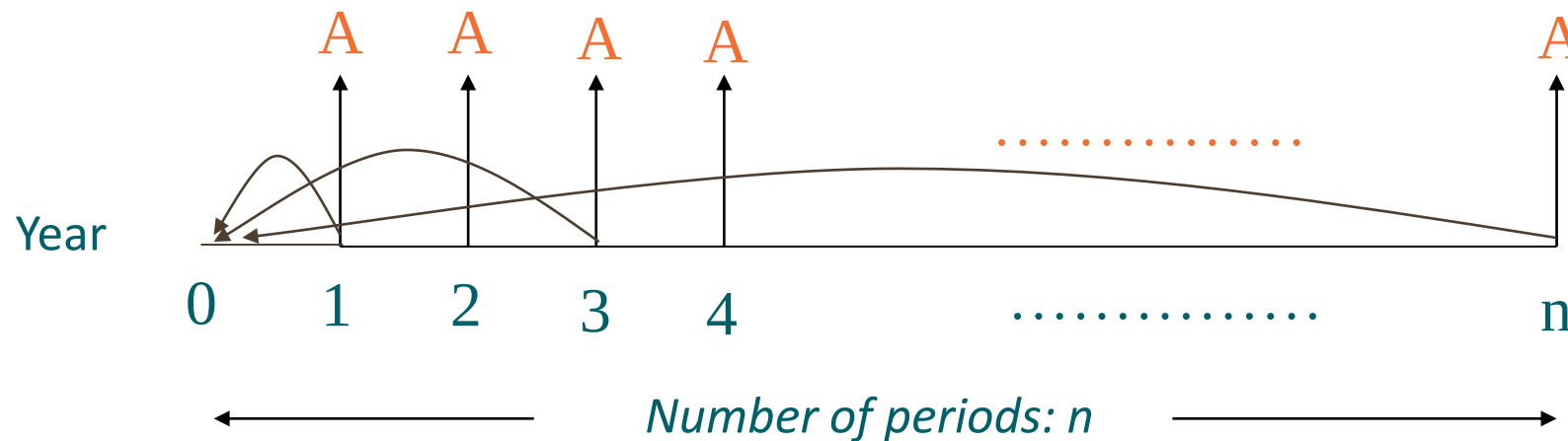
$$X.PV(P) = aX + aX^2 + aX^3 + \dots + aX^\infty \quad (2)$$

Subtracting the former equation from the latter one, we get:

$$\mathbf{PV(P) = P/r}$$

The present value of an annuity

- ◆ What is an annuity?
 - Denote an annuity as A
 - An annuity is the equal net benefit that lasts from year 1 to year n .
 - It can be illustrated as:



The present value of an annuity

- It is also defined by the difference b/w two perpetuities:

Year	0	1	...	n	n+1	...	∞
P_1		A		A	A	...	A
P_2		0		0	A	...	A
A		A	...	A			

$$PV(P_1) = \frac{A}{r}$$

$$PV(P_2) = \frac{A}{r} \left[\frac{1}{(1+r)^n} \right]$$

$$\Rightarrow A = P_1 - P_2$$



$$PV(A) = PV(P_1) - PV(P_2)$$

$$= \frac{A}{r} - \frac{A}{r} \left[\frac{1}{(1+r)^n} \right]$$

$$= A \left[\frac{1}{r} - \frac{1}{r} \cdot \left(\frac{1}{(1+r)^n} \right) \right]$$

$$AF = \left[\frac{1}{r} - \frac{1}{r(1+r)^n} \right]$$

$$AF = \left[\frac{(1+r)^n - 1}{r(1+r)^n} \right]$$

$$AF = \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

AF: Annuity Factor.

The present value of an annuity

Another way to derive the AF formula:

- ◆ An annuity consisting of \$1 to be paid one year from now and in every subsequent year until n has a present value:

$$S = \frac{1}{(1+r)} + \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \frac{1}{(1+r)^4} + \dots + \frac{1}{(1+r)^n} \quad (1)$$

- ◆ If we multiply both sides of this equation by $(1+r)$, we get:

$$S(1+r) = 1 + \frac{1}{(1+r)} + \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \frac{1}{(1+r)^4} + \dots + \frac{1}{(1+r)^{n-1}} \quad (2)$$

- ◆ Subtracting the former equation from the latter gives the above formulae.

The present value of an annuity

- ♦ What is the feature of the annuity factor, (AF_r^n)?

$$AF = \frac{1}{(1+r)^1} + \frac{1}{(1+r)^2} + \dots + \frac{1}{(1+r)^n}$$

- ♦ Some applications of AF_r^n :

- PMT formula
- Quick calculations of various annuities such as rent, insurance, bond valuation, ... (by using the **PV** function)
- Derive annual equivalent values of projects with different life's horizons
- Derive the compound factor of an annuity (by using the **FV** function):

$$AF_r^n (1+r)^n = \left[\frac{(1+r)^n - 1}{r(1+r)^n} \right] (1+r)^n = \left[\frac{(1+r)^n - 1}{r} \right] = CF_r^n$$

- ◆ Calculate the total present value of an annuity in Excel:

=PV(rate,nper,-pmt)

- ◆ For example, a project invests 10000 (NCF_0), and generates an annual NCF (from year 1) of 2000; discount rate is 15%, and the project's life is 15 years, then its NPV is:

$$=PV(15\%,15,-2000) - 10000 = 38$$

- ◆ Calculate the annual equivalent values or annual debt repayments in Excel:

=-PMT(rate,nper,PV)

- ◆ For example, a project borrows 5000 at year 0 with an interest rate of 8% per annum, and the repayment term is 5 years, then the equal repayments (principal plus interest charge) are:

$$=-PMT(8\%,5,5000) = 1252$$

- ◆ Calculate the total future value of an annuity in Excel:

=FV(rate,nper,-pmt)

- ◆ For example, you open a savings account with a fixed deposit of 10 million per month for 12 months with an interest rate of 0.8% per month, you will earn a total value:

$$=FV(0.8\%,12,-10) = 125.4 \text{ million}$$

Alternative investment criteria

Net present value (NPV)



Net present value (NPV)

- Let's denote:

NPV : Net present value

NCF_t : Net cash flow

t : $0, 1, 2, \dots, n$

r : Constant discount rate over the project's life.

$$NPV = \sum_{t=0}^n \frac{NCF_t}{(1+r)^t}$$

- Formula:

$$NPV = NCF_0 + \frac{NCF_1}{(1+r)^1} + \frac{NCF_2}{(1+r)^2} + \dots + \frac{NCF_n}{(1+r)^n}$$

Net present value (NPV)

- Financial function of the NPV formula?

- Any cash flow:

$$=NPV(r, NCF_1:NCF_n) + NCF_0$$

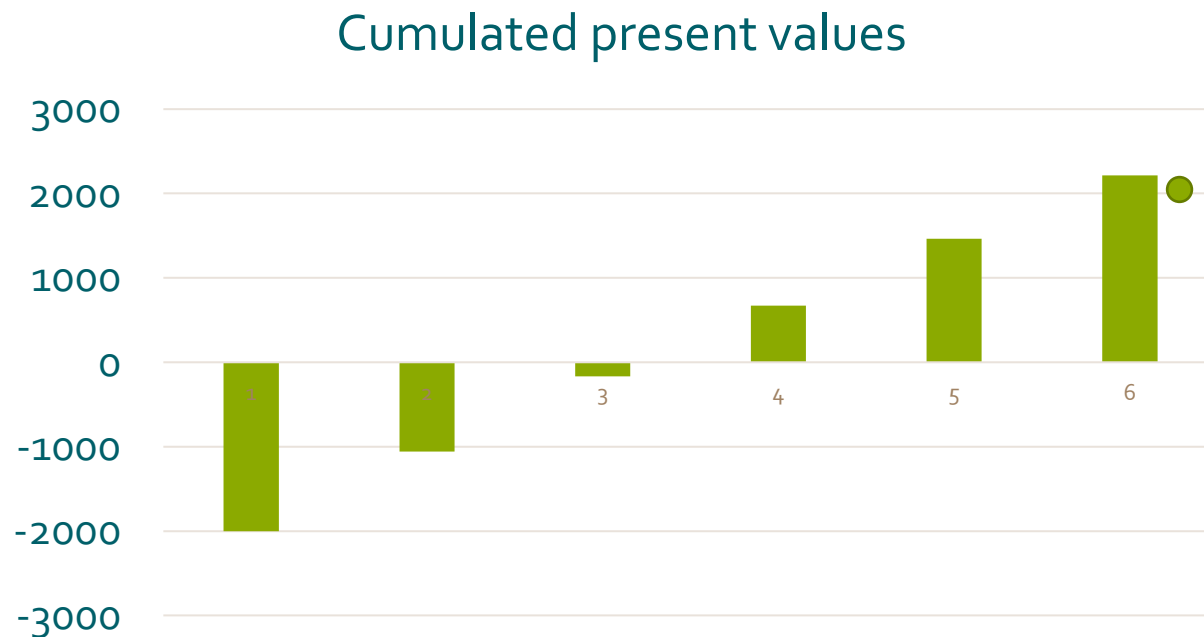
- Equal cash flow, i.e. $NCF_1 = NCF_2 = \dots = NCF_n = A$
(or an annuity):

$$=PV(r, nper, -A) + NCF_0$$

	A	B	C	D	E	F	G	H
14		Year	0	1	2	3	4	5
15		1. Receipts		0	3247	4573	3525	2338
16		2. Expenditures	5000	2120	1000	1000	1000	1000
17		3. Net cash flows	-5000	-2120	2247	3573	2525	1338
18		4. Discount factors @6%	1.00	0.94	0.89	0.84	0.79	0.75
19		5. Present values [=4*3]	-5000	-2000	2000	3000	2000	1000
20		6. Net present value [NPV]	1000					
21		NPV function in Excel: =NPV(6%,D17:H17)+C17						
22		NPV(rate, value1, [value2], [value3], ...)						

	A	B	C	D	E	F	G	H
31		Year	0	1	2	3	4	5
32		1. Net cash flow	-2000	1000	1000	1000	1000	1000
33		2. Discount factors @ = 6%	1.00	0.94	0.89	0.84	0.79	0.75
34		3. Present values [=1*2]	-2000	943	890	840	792	747
35		4. Net present value [NPV]	2212					
36		PV function in Excel: =PV(6%,5,-1000)+C34						
37		PV(rate, nper, pmt, [fv], [type])						

	A	B	C	D	E	F	G	H
38		Cumulated present values						
39		Year	0	1	2	3	4	5
40		1. Net cash flow	-2000	1000	1000	1000	1000	1000
41		2. Discount factors @ = 6%	1.00	0.94	0.89	0.84	0.79	0.75
42		3. Present values [=1*2]	-2000	943	890	840	792	747
43		4. Cumulated present values	-2000	-1057	-167	673	1465	2212



This is
the NPV

	A	B	C	D	E	F	G	H	I	J
2	Does zero make a difference?									
3										
4		Year	0	1	2	3	4			
5		NCF_A	-1000	0	500	500	500			
6		NCF_B	-1000		500	500	500			
7		DF_A	1	0.87	0.76	0.66	0.57			
8		DF_B	1		0.87	0.76	0.66			
9		Discount rate	15%							
10		NPV_A	-7.29	-7.29	Correct					
11		NPV_B	142	142	Incorrect					
12										
13										
14										

Answer: NO.
The NPV function in Excel automatically assumes the 'blank' cell is Year 0.

If a NCF is zero, can we keep that cell blank?

Net present value (NPV)

- If a private company undertakes the project, the financial NPV is the expected added value of the company.

$$\text{Value of a Firm} = \sum \text{PV of Existing Projects} + \sum \text{NPV of Expected Future Projects}$$

- The NPV formula implicitly takes into account the re-invested rate of the positive net cash flows of the project.

This re-invested rate is equal the discount rate.

Suppose the discount rate is 10%.

Year	0	1	2	3
NCF	-300	100	200	300
FV(NCF) _{10%}		121	220	300
Total FV				641
PV(641) _{10%}	482			
$\text{NPV} = 482 - 300 = 182$				
NPV, Excel function = 182				

Net present value (NPV)

- The NPV formula is extended to the case that the discount rates will vary from period to period over the life of the project.

$$NPV = NCF_0 + \frac{NCF_1}{(1 + r_1)} + \frac{NCF_2}{(1 + r_1)(1 + r_2)} + \frac{NCF_3}{(1 + r_1)(1 + r_2)(1 + r_3)} + \dots \frac{NCF_n}{(1 + r_1)(1 + r_2) \dots (1 + r_n)}$$

- NPV can be additive.

Year	0	1	2	NPV (10%)
Net cash flow of project 1	-100	0	550	354
Net cash flow of project 2	-100	225	0	105
Net cash flow of project 3	-100	450	0	309
Net cash flow of projects 1 & 3	-200	450	550	663
Net cash flow of projects 2 & 3	-200	675	0	414

Net present value (NPV)

- **Rule 1:**

Do not accept any project unless it generates a positive NPV when discounted by the opportunity cost of funds.

- **Rule 2:**

Within the limit of a fixed budget, choose the subset of the available projects that maximizes the NPV.

- **Rule 3:**

When there is no budget constraint but a project must be chosen from mutually exclusive alternatives, investors should always choose the alternative that generates the largest net present value.

With a fixed budget for capital expenditures of \$4 million, which projects should be chosen?

	<u>Project A</u>	<u>Project B</u>	<u>Project C</u>	<u>Project D</u>
PV investment costs	\$1.0 million	\$3.0 million	\$2.0 million	\$2.0 million
NPV of net benefits	+\$60,000	+\$400,000	+\$150,000	+\$225,000

Three projects, E, F and G that are mutually exclusive for technical reasons. Which one should be chosen?

	<u>Project E</u>	<u>Project F</u>	<u>Project G</u>
PV investment costs	\$1.0 million	\$4.0 million	\$1.5 million
NPV of net benefits	+\$300,000	+\$700,000	+\$600,000

Net present value (NPV)

■ Problem #1 of the NPV criterion:

If we compare the NPV of projects with different lifetimes, the results could be misleading.

- How to overcome this problem?
 - Repeat shorter lifetime projects through time
 - Apply the annual equivalent method
 - Use adjusted coefficient

COMPARE PROJECTS WITH DIFFERENT LIFETIMES

	Projects (mutually exclusive)	Initial investment (year 0)	Annual NCF (from year 1)	Lifetime	NPV	AE
NPV	A	75000	23000	6	12043	3182
	B	55000	20000	5	12043	3593
	C	40000	17500	4	9962	3489
IRR	A	21%	21%	21%		
	B	24%	24%	24%		
	C	27%	27%	27%		

With the AE, project B becomes the best option; and project C is even better than project A.

When projects or alternatives have different lifetimes, we should calculate the NPV of each project, then use the PMT function to convert the NPV values to the annual equivalent benefits or costs. This method provides a better criterion for comparison. DON'T USE EITHER NPV OR IRR CRITERIA.

If the NPV is used, projects A and B look better than project C.

Alternative investment criteria

Internal rate of return (IRR)



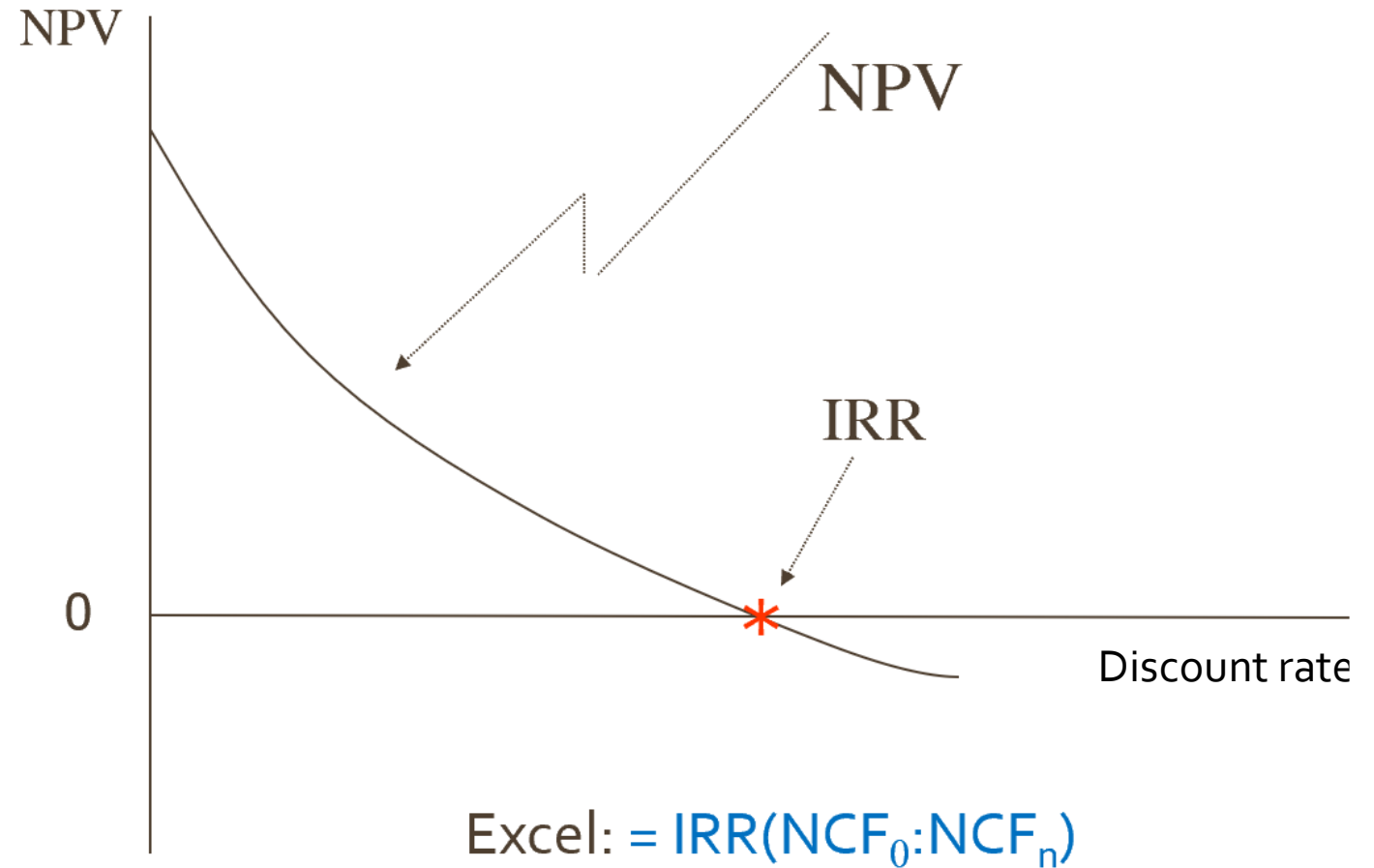
Internal rate of return (IRR)

- ◆ The internal rate of return (IRR) for a project is the discount rate that is obtained by the solution of the following equation:

$$NPV = \sum_{t=0}^n \frac{NCF_t}{(1+r^*)^t} = 0$$

=> r^* is IRR

- ◆ IRR is the average rate of return of the project.



Internal rate of return (IRR)

$$NPV = \sum_{t=0}^n \frac{CIF_t}{(1+k)^t} - \sum_{t=0}^n \frac{COF_t}{(1+k)^t}$$

$$NPV = 0 = \sum_{t=0}^n \frac{CIF_t}{(1+IRR)^t} - \sum_{t=0}^n \frac{COF_t}{(1+IRR)^t} \quad \sum_{t=0}^n \frac{COF_t}{(1+IRR)^t} = \sum_{t=0}^n \frac{CIF_t}{(1+IRR)^t}$$

$$\sum_{t=0}^n \frac{COF_t}{(1+IRR)^t} = \sum_{t=0}^n \frac{CIF_t (1+IRR)^n}{(1+IRR)^t (1+IRR)^n} = \frac{\sum_{t=0}^n CIF_t (1+IRR)^{n-t}}{(1+IRR)^n}$$

Internal rate of return (IRR)

- **Rule 1:**

Do not accept a project unless its IRR is higher than the opportunity cost of funds.

- **Rule 2:**

When there is no budget constraint but a project must be chosen from mutually exclusive alternatives, investors should choose the alternative that generates the highest internal rate of return.

Internal rate of return (IRR)

- The IRR criterion has seen considerable use by both private and public sector investors as a way of describing the attractiveness of a particular project.
- However, it is not a reliable investment criterion as there are **several problems** associated with it.

Problem #1: The IRR may not be unique.

- For example, a project invests $K = -1.6$, generates net benefits $B_1 = 10$ & $B_2 = 10$, an internal rate of return $r_p > 0$ is the root of this equation:

$$-K + \frac{B_1}{(1+r_p)} + \frac{B_2}{(1+r_p)^2} = 0$$

- Multiply both sides by $(1+r_p)^2$ and denote $x = (1+r_p)$, we get:

$$-Kx^2 + B_1x + B_2 = 0$$

- Let $a = -1.6$, $b = 10$ & $c = 10$, we have the following equation:

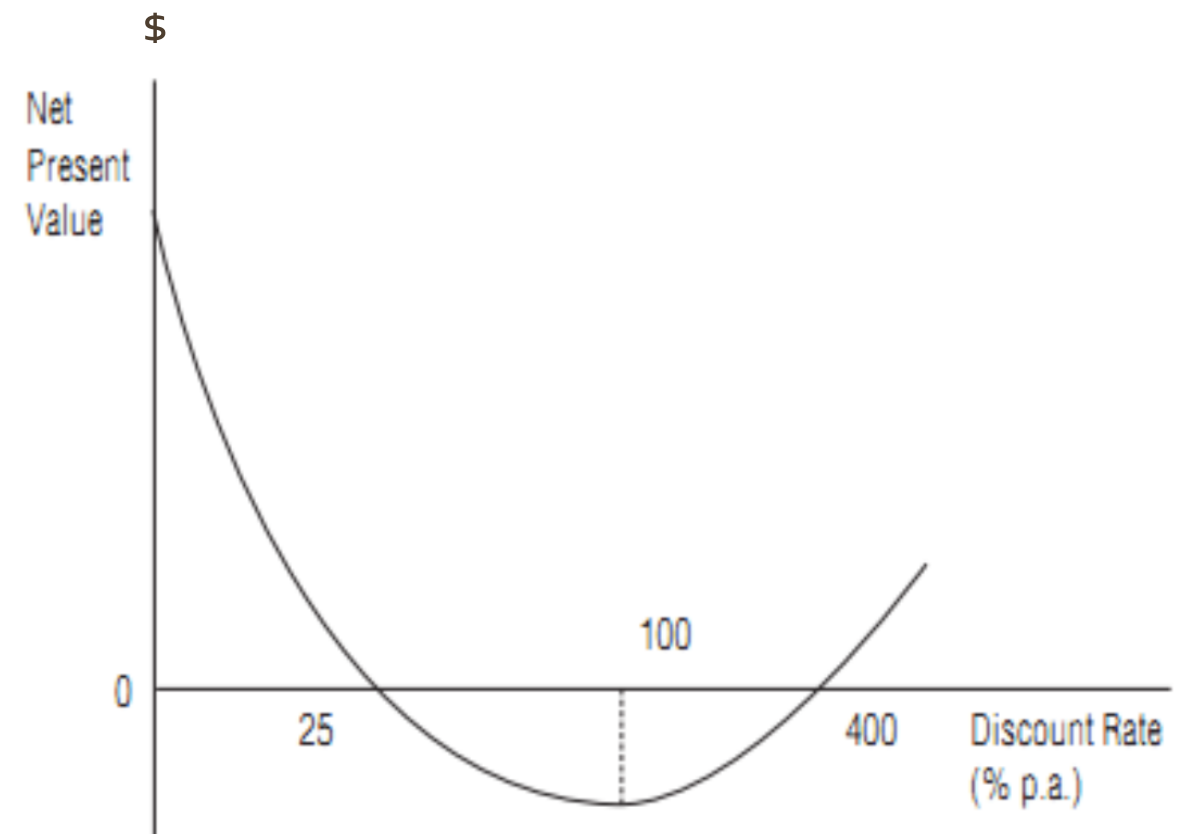
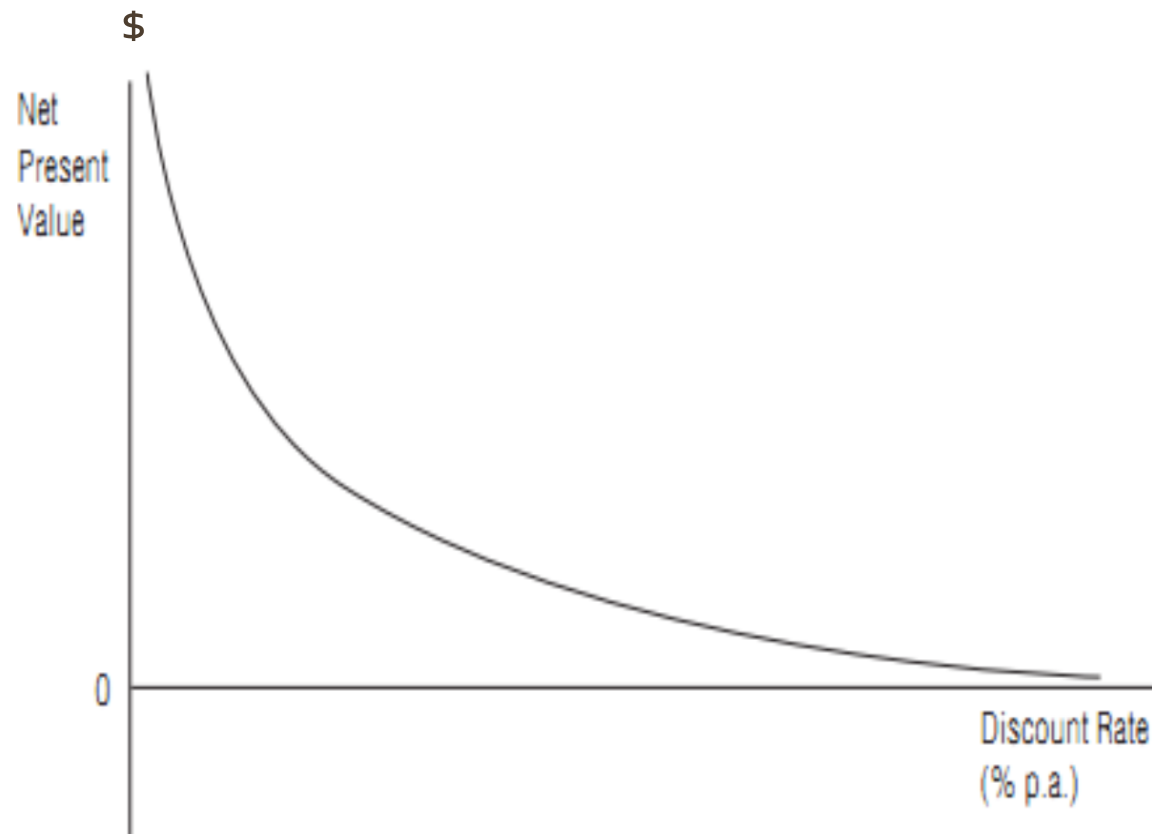
$$F(x) = ax^2 + bx + c = 0$$

- So:

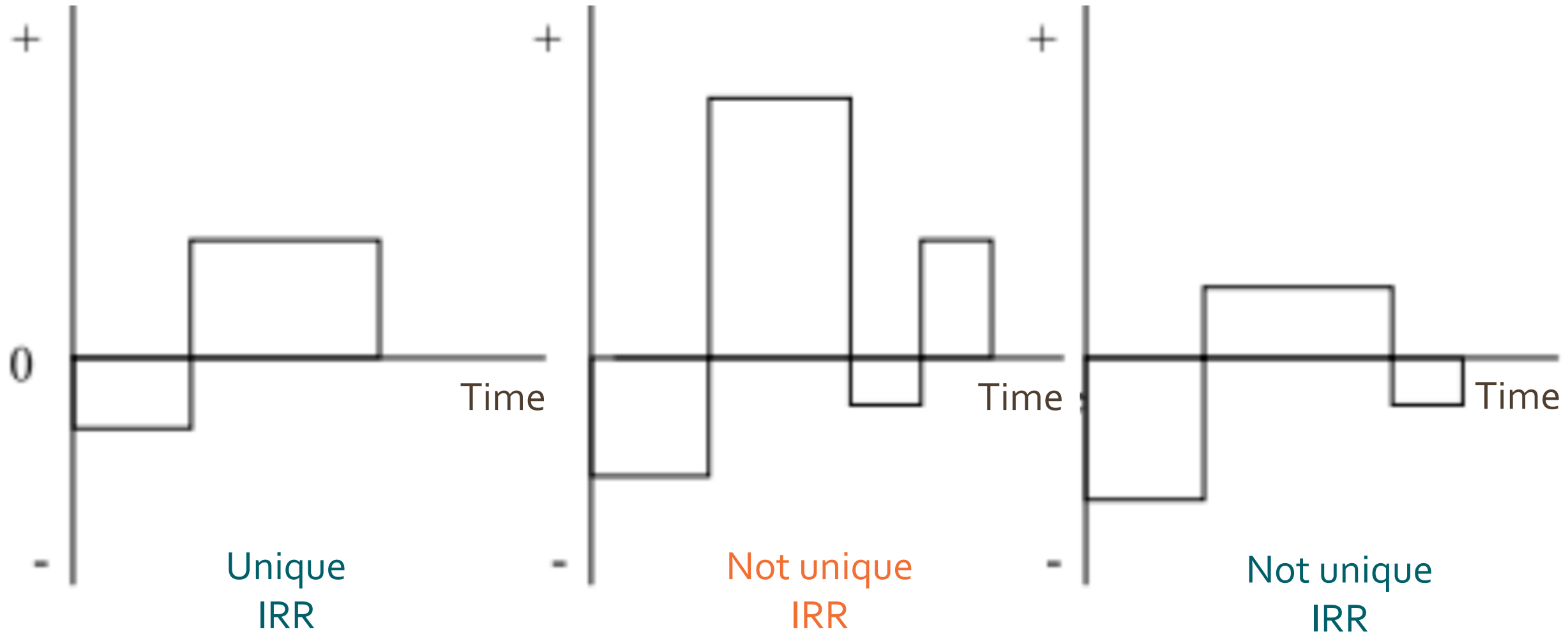
$$x = -b \pm \sqrt{\frac{(b^2 - 4ac)}{2a}} \quad \longrightarrow \quad r_p = -188\% (< 0, \text{discarded}), \text{ and } r_p = 613\%$$

- If we change $B_2 = -10$, we have two positive solutions: $r_p = 25\%$ and $r_p = 400\%$
- If we change $K = 1.6$, there is no value of r_p b/c $F(x)$ now consists of three positive values.

When the net benefits have the same signs (all + or all –), there is no IRR; when the net benefits change signs once - +, there is one IRR; when the net benefits change signs two times, - + -, there are two values of IRR; etc.

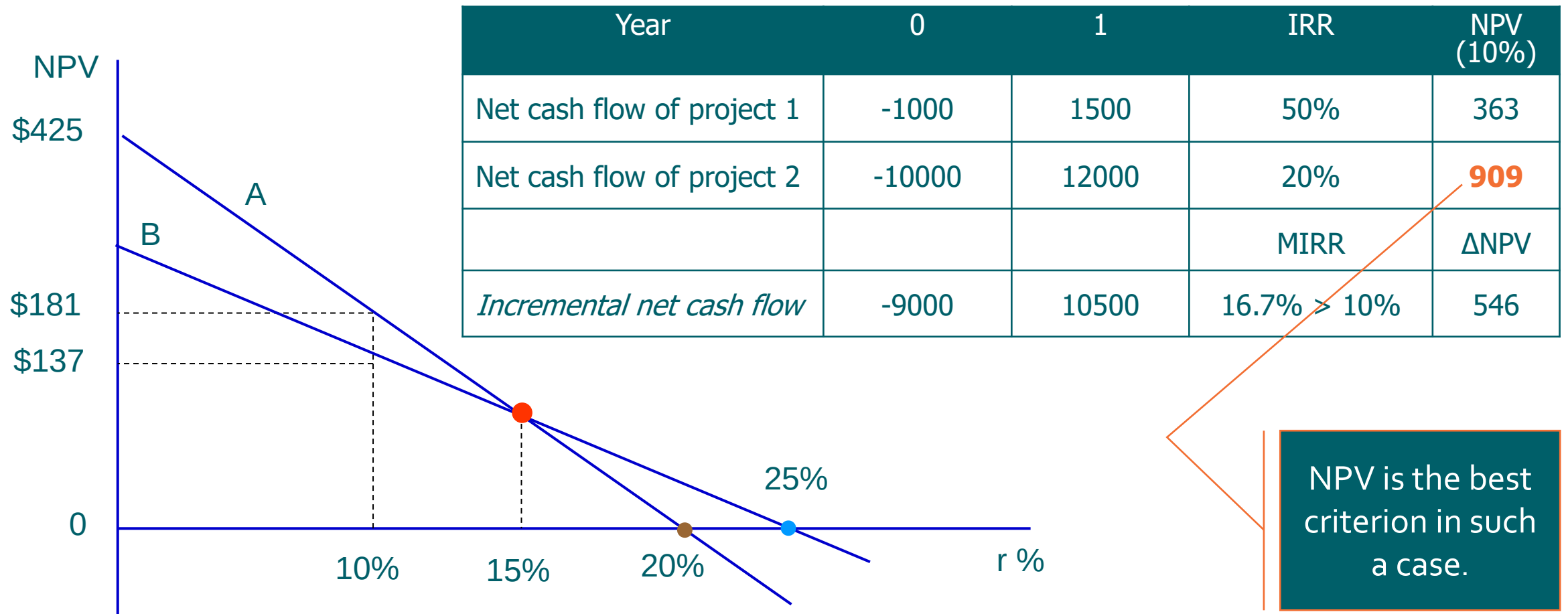


Different project net cash flows



Problem #2: Mutually exclusive projects of different scale.

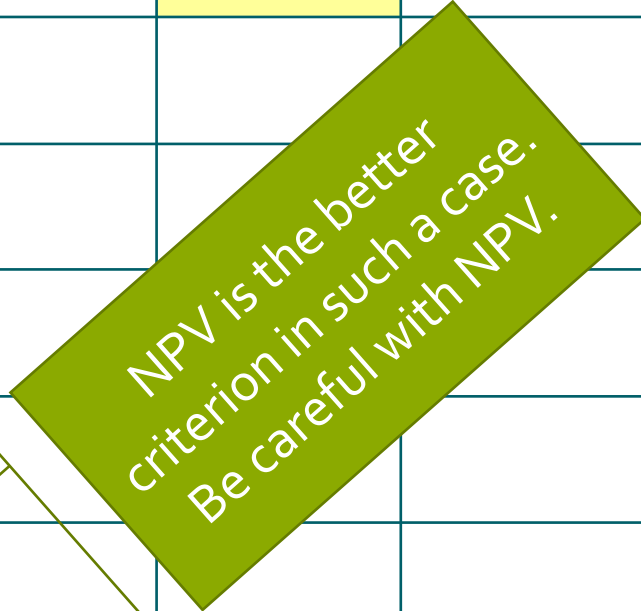
Whereas the NPV takes explicit account of the scale of the project by means of the investment that is required, the IRR ignores the differences in scale.



Problem #3: Projects with different lengths of life.

The IRR criterion is unreliable for project selection when alternative projects have different lengths of life. It favors projects with shorter lives.

Year	0	...	5	...	10
Net cash flow of project 1	-1000		3200		
Net cash flow of project 2	-1000				5200
NPV(1, 8%) = 1.178					
NPV(2, 8%) = 1.409					
IRR ₁ = 26.2%					
IRR ₂ = 17.9%					



NPV is the better criterion in such a case.
Be careful with NPV.

Problem #4: Projects with different timing.

The IRR criterion can cause us to make the incorrect choice of projects with different timing.

NPV is the best criterion in such a case.

Year	0	1	2	3	IRR	NPV (8%)
Net cash flow of project 1	-1000	1400			40%	296
Net cash flow of project 2		-1000	1400		40%	274
Net cash flow of project 3	-1000	1500			50%	389
Net cash flow of project 4			-1100	1670	52%	383

Problem #5: Do not distinguish borrowing and lending projects.

With a discount rate of 15% per annual, the IRR criterion can not distinguish between the lending versus borrowing projects.

Year	0	1	IRR	NPV
Lending net cash flow (Bank)	-1000	1200	20%	43.48

Year	0	1	IRR	NPV
Borrowing net cash flow (Equity)	1000	-1200	20%	-43.48

Problem #6: The IRRs are not additive.

Year	0	1	2	IRR	NPV (10%)
Net cash flow of project 1	-100	0	550	135%	354
Net cash flow of project 2	-100	225	0	125%	105
Net cash flow of project 3	-100	450	0	350%	309
Net cash flow of projects 1 & 3	-200	450	550	213%	663
Net cash flow of projects 2 & 3	-200	675	0	238%	414

Problem #7: Irregularity of cash flows.

Year	0	1	2	3	4	IRR
Project A	1,000	1,200	800	3,600	-8,000	10%
Project B	1,000	1,200	800	3,600	-6,400	-2%
Project C	1,000	1,200	800	3,600	-4,800	-16%
Project D	-1,000	1,200	800	3,600	-4,800	4%
Project E	-1,325	1,200	800	3,600	-4,800	20%

Results:

Project B is obviously better than project A, yet $IRR_A > IRR_B$.

Project C is obviously better than project B, yet $IRR_B > IRR_C$.

Project D is worse than project C, yet $IRR_D > IRR_C$.

Project E is worse than project D, yet $IRR_E > IRR_D$.

Source: Jenkins et al. (Chapter 4, P.17)

Problem #8: Re-invested rate is equal IRR of the project.

The IRR formula also takes into account the re-investment rate of the positive net cash flows. However, the re-investment rate is sometimes not appropriate. Therefore, we should use the modified IRR (MIRR) instead.

Year	0	1	2	3
Cash inflows (CIF)			150	200
Cash outflows (COF)	100	110		
NCF = CIF – COF	-100	-110	150	200
PV(COF) at r = 10%	200			
FV(CIF) in year 3 at r=10%				365

The problem is to find a discount rate to convert the future value of 365 in year 3 to its equivalent value that equals the present value of cash outflow, 200 at year 0. Such a discount rate is known as MIRR criterion.

Re-invested rate

k could be either WACC, r_e , r_d or any value depending on the viewpoints of appraisal.

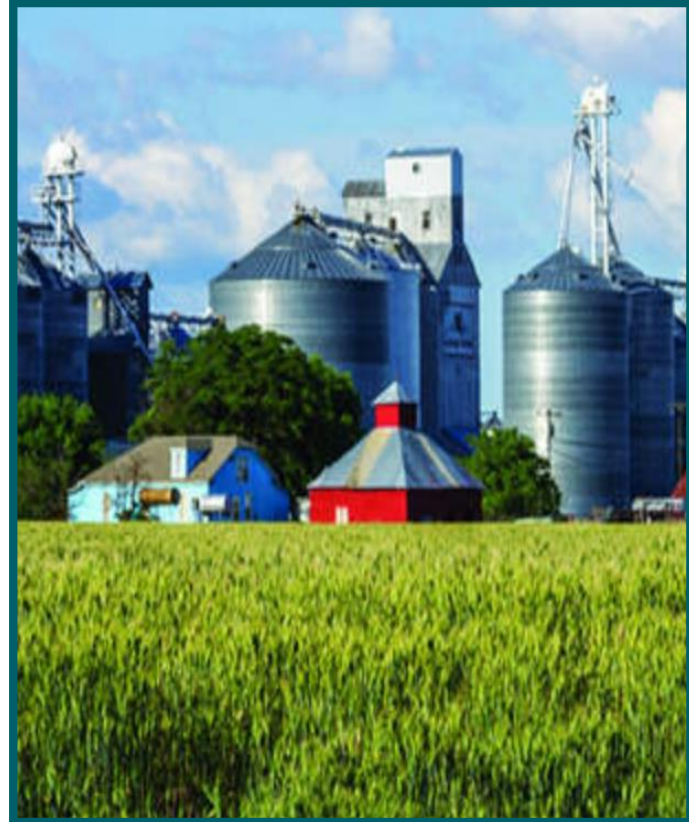
$$\sum_{t=0}^n \frac{\text{COF}_t}{(1 + \mathbf{r})^t} = \frac{\sum_{t=0}^n \text{CIF}_t (1 + \mathbf{k})^{n-t}}{(1 + \mathbf{MIRR})^n}$$

Finance rate

Excel: =MIRR(NCF₀:NCF_n, r, k)

Alternative investment criteria

Cost to benefit ratio (BCR)



Cost to benefit ratio (BCR)

$$BCR = \frac{NPV(CIF)}{NPV(COF)}$$

r is the opportunity cost of funds or the discount rate.

$$BCR = \frac{\sum_{t=0}^n \frac{CIF_t}{(1+r)^t}}{\sum_{t=0}^n \frac{COF_t}{(1+r)^t}}$$

Rule 1:

Do not accept a project unless its BCR is greater than 1.

Rule 2:

When a choice must be made between two or more mutually exclusive projects, investors should select the project with the higher, or highest, BCR.

Problem #1: BCR is sensitive to the way in which costs are defined.

Problem #2: BCR criterion would lead to an incorrect decision of mutually exclusive projects with different scale.

Project	A	B
1. PV of cash inflows	2000	2000
2. PV of initial investment costs	1200	100
3. PV recurrent costs	500	1800
4. NPV of project	300	100
$BCR_1 = (1 - 3)/2$	1.25	2.0
$BCR_2 = 1/(2 + 3)$	1.18	1.05

PV(NCF)/
PV(Initial
investment costs)

PV(Cash inflows)/
PV(Cash outflows)

NPV is still the best criterion!

BCR ranks
smaller scale
projects
higher.

Project	PV of investment cost	PV of cash inflows	NPV of project	BCR
C	1000	1300	300	1.3
D	8000	9400	1400	1.18
E	1500	2100	600	1.4

Problem #2 of NPV

	<i>Year</i>					<i>IRR</i>	<i>NPV</i>
	<i>0</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>4%</i>	<i>10%</i>
A	-1000	500	500	500	23%	387	243
B	-2362	1000	1000	1000	13%	413	125

- $IRR_A > IRR_B \Rightarrow$ Project A is better than project B.
- However, if we use a low discount rate ($r = 4\%$), then $NPV_B > NPV_A \Rightarrow$ Project B is better than project A.
- Conversely, if we use a high discount rate ($r = 10\%$), then $NPV_A > NPV_B \Rightarrow$ Project A is better than project B.
- Try using the IRR of the incremental cash flow ($B - A$) as a decision rule in this example.

Problem#3 of NPV

Project	PV(K)	PV(NB)	NPV (rank)	NB/K (rank)	PV(K) = Present value of the initial investment outlay
A	100	130	30 (5)	1.30 (2)	PV(NB) = Present value of the net benefits
B	400	433	33 (4)	1.08 (5)	NPV = Net Present Value (ranked)
C	200	303	103 (1)	1.52 (1)	$\frac{NB}{K}$ = Ratio of PV(NB) over PV(K) (ranked)
D	400	494	94 (2)	1.24 (3)	
E	500	558	58 (3)	1.11 (4)	

- As the budget is limited to 800 and assuming that E is a divisible investment project.
- Using the the NPV rule for ranking, the best combination is C + D + 0.4E => the total NPV obtained from this package would be 220.
- Using the the BCR rule for ranking, the best combination is C + A + D + 0.2E => the total NPV obtained from this package would be 239.
- BCR is clearly a better way of ranking potential investments in situations of capital rationing, in comparison with ranking according to NPVs.

Alternative investment criteria

Pay-out or pay-back period



Payback period (PP)

- The pay-out or pay-back period measures the number of years it will take for the net cash flows to repay the capital investment.
- The criterion puts a large premium on projects which have a quick pay-back and thus it has been a popular criterion in making business investment choices.
- However, it may provide the wrong results especially in the cases of investments with a long life where future net benefits are known with a considerable degree of certainty.

Payback period (PP)

There are two versions of the pay-back period criterion:

- It uses the undiscounted net cash flows to repay the capital investment;
- It compares the discounted benefits over a number of years from the beginning of the project with the discounted investment costs.

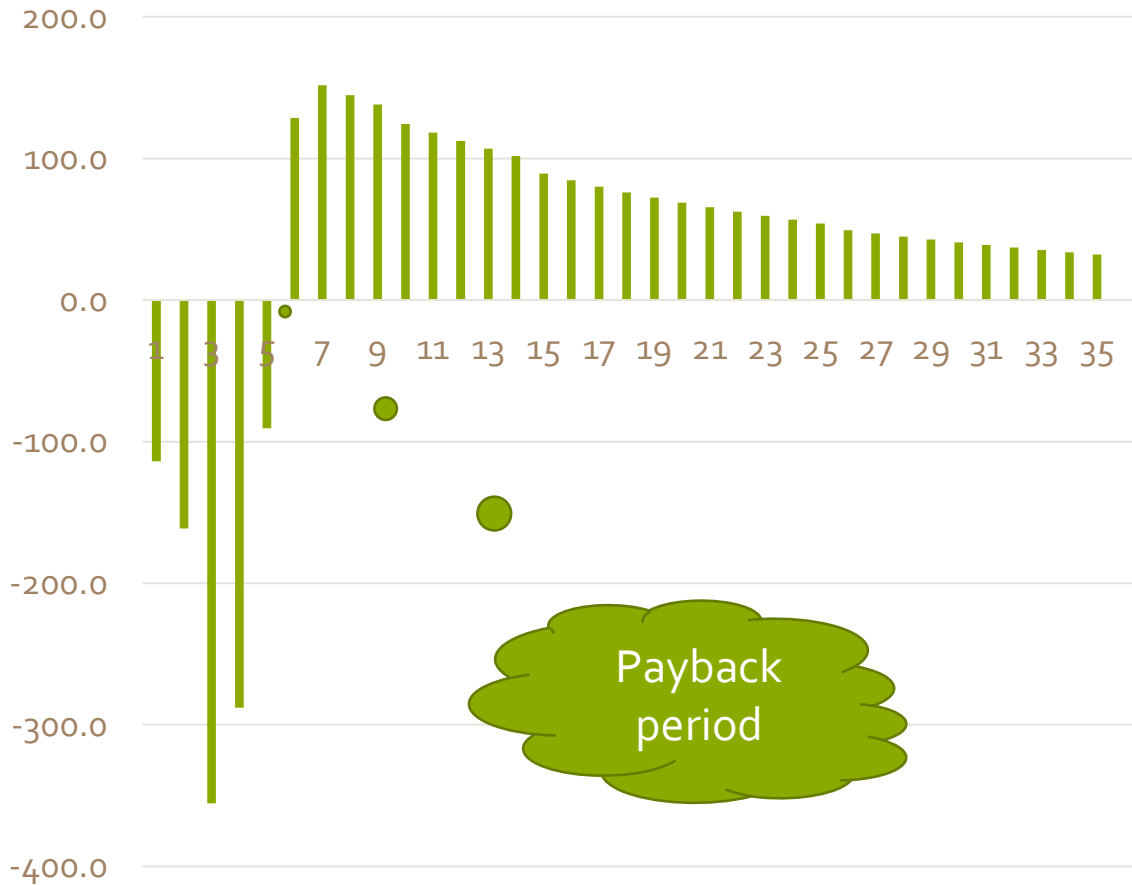
Steps in calculating the discounted pay-back period:

- 1) Calculate $PV(NCF)$
- 2) Derive the cumulated $PV(NCF)$
- 3) Determine the year (y^*) which is the year before the cumulated $PV(NCF)$ is positive
- 4) Determine the month m^* by using the proportional rule, that is the absolute value of $PV(NCF_{y^*}) \times 12 / PV(NCF_{(y^*+1)})$.

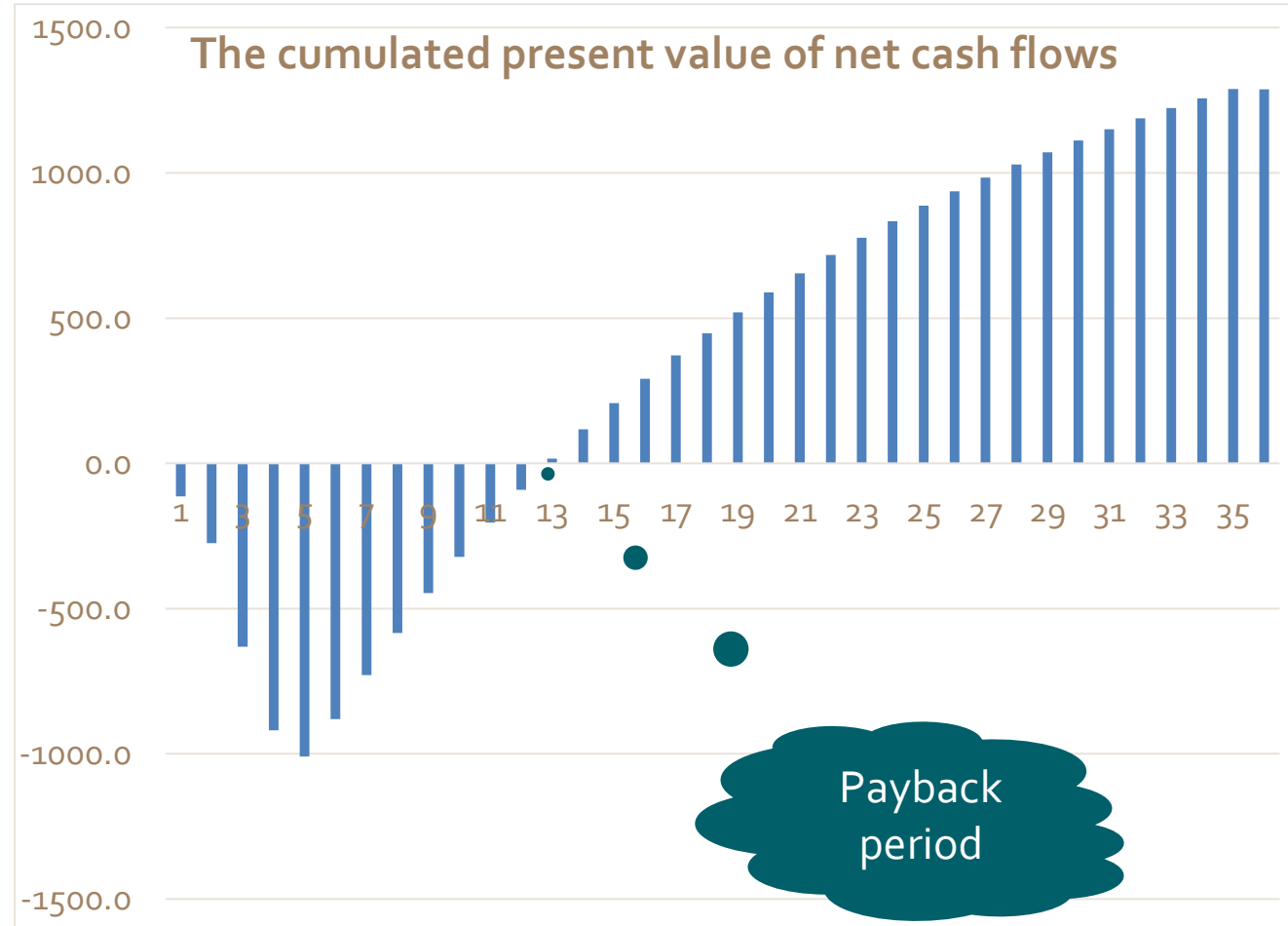
$$PP = y^* + m^*$$

The PP with discounted cash flows

The present value of net cash flows



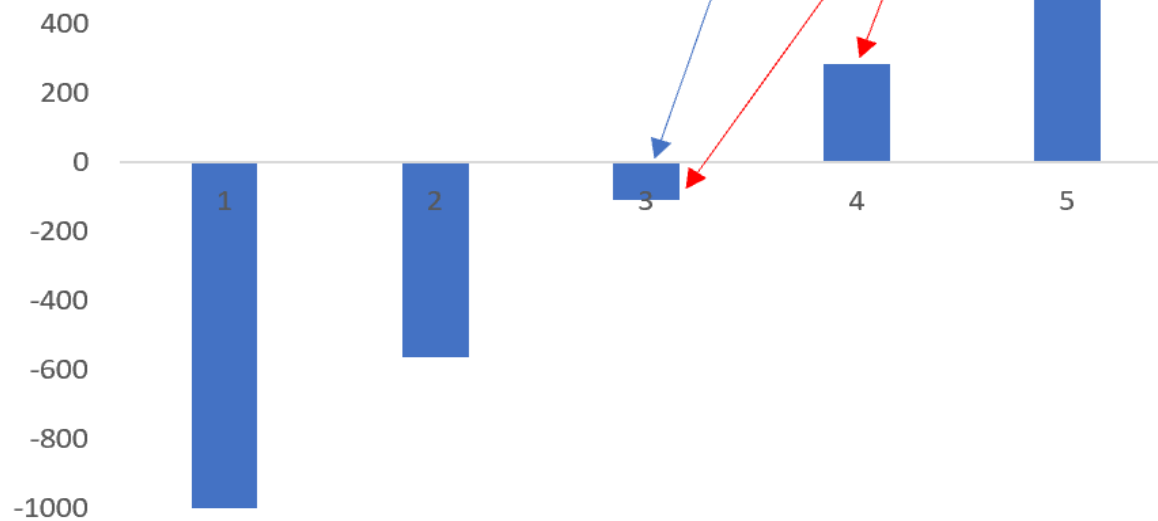
The cumulated present value of net cash flows



PAYBACK PERIOD

Year	0	1	2	3	4
NCF	-1200	500	600	600	400
Cumulated NCF	-1200	-700	-100	500	900
			2 years	2 months	
Year	0	1	2	3	4
NCF	-1000	500	600	600	400
DF	1.000	0.870	0.756	0.658	0.572
PV	-1000	435	454	395	229
Cumulated PV	-1000	-565	-112	283	512
			2 years	3.4	months

Cumulated present values



NPV

Financial sustainability

Debt service coverage ratios



Financial sustainability

- The project is financially sustainable when the risk of running out of cash in the future, both during the investment and the operating stages, is expected to be nil.
- The cumulated cash flow should be zero (or positive) during the construction phase.
- The cumulated net cash flow is greater than zero for all the years considered.

	Years							
	1	2	3	4	5-9	10	11-29	30
Sources of financing	8,465	75,176	42,890					
Total revenues				11,598	...	12,011	...	12,222
Total inflows	8,465	75,176	42,890	11,598	...	12,011	...	12,222
Initial investment	8,465	75,176	42,890					
Replacement costs						11,890	9,760	
Loan repayment (including interest)					1,789	1,789	1,789	
Total operating costs				5,561	...	5,662	...	5,713
Taxes				604	...	-733	...	651
Total outflows	8,465	75,176	42,890	5,561	...	19,341		5,713
Net cash flow	0	0	0	6,037	...	-7,329	...	6,509
Cumulated net cash flow	0	0	0	6,037	...	20,726	...	133,835

The cumulated cash flow should be zero (or positive) during the construction phase

Financial sustainability is verified if the cumulated net cash flow row is greater than zero for all the years considered.

Debt service coverage ratio (DSCR)

- The debt service coverage ratio is a key factor in determining the ability of a project to pay its operating expense and to meet its debt servicing obligations.
- This is used by bankers who want to know annual debt service coverage ratio of a project on a year-to-year basis as well as a summary of the loan life cover ratio.

$$ADSCR_t = [ANCF_t / (\text{Annual Debt Repayment}_t)]$$

Annual debt service capacity ratio

$$LLCR_t = PV(ANCF_{t \text{ to end year of debt}}) / PV(\text{Annual Debt Repayment}_{t \text{ to end year of debt}})$$

Loan life cover ratio

The discount rates used are the interest rate being paid on the loan financing.

Year	0	1	2	3	4	5	6	7	8	9	10
Net Cash Flow	-2,000,000	320,000	320,000	360,000	440,000	380,000	100,000	200,000	480,000	540,000	640,000
Debt Repayment	0	298,316	298,316	298,316	298,316	298,316					
ADSCR		1.07	1.07	1.21	1.47	1.27					

Decrease the interest rate on the loan.

Year	0	1	2	3	4	5	6	7	8	9	10
Net Cash Flow	-2,000,000	320,000	320,000	360,000	440,000	380,000	100,000	200,000	480,000	540,000	640,000
Debt Repayment	0	206,040	206,040	206,040	206,040	206,040					
ADSCR		1.55	1.55	1.75	2.14	1.84					

Decrease the amount of debt financing.

Year	0	1	2	3	4	5	6	7	8	9	10
Net Cash Flow	-2,000,000	320,000	320,000	360,000	440,000	380,000	100,000	200,000	480,000	540,000	640,000
Debt Repayment	0	178,989	178,989	178,989	178,989	178,989					
ADSCR		1.79	1.79	2.01	2.46	2.12					

Increase the duration of loan repayment.

Year	0	1	2	3	4	5	6	7	8	9	10
Net Cash Flow	-2,000,000	320,000	320,000	360,000	440,000	380,000	100,000	200,000	480,000	540,000	640,000
Debt Repayment	0	199,252	199,252	199,252	199,252	199,252	199,252	199,252	199,252	199,252	199,252
ADSCR		1.61	1.61	1.81	2.21	1.91	0.50	1.00	2.41	2.71	3.21

Ask for the bridge financing.

Year	0	1	2	3	4	5	6	7	8	9	10
Net Cash Flow	-2,000,000	320,000	320,000	360,000	440,000	380,000	100,000	200,000	480,000	540,000	640,000
Debt Repayment	0	199,252	199,252	199,252	199,252	199,252	199,252	199,252	199,252	199,252	199,252
ADSCR		1.61	1.61	1.81	2.21	1.91	0.50	1.00	2.41	2.71	3.21
NPV of NCF		2,052,134	1,991,954	1,922,747	1,797,159	1,560,733	1,357,843	1,446,519	1,433,497	1,096,522	640,000
PV of Debt Repayments		1,150,000	1,093,360	1,028,224	953,318	867,176	768,112	654,189	523,178	372,515	199,252
LLCR		1.78	1.82	1.87	1.89	1.80	1.77	2.21	2.74	2.94	3.21

An illustrative example

ICP Project, Campbell & Brown (2016, chapters 4, 13)

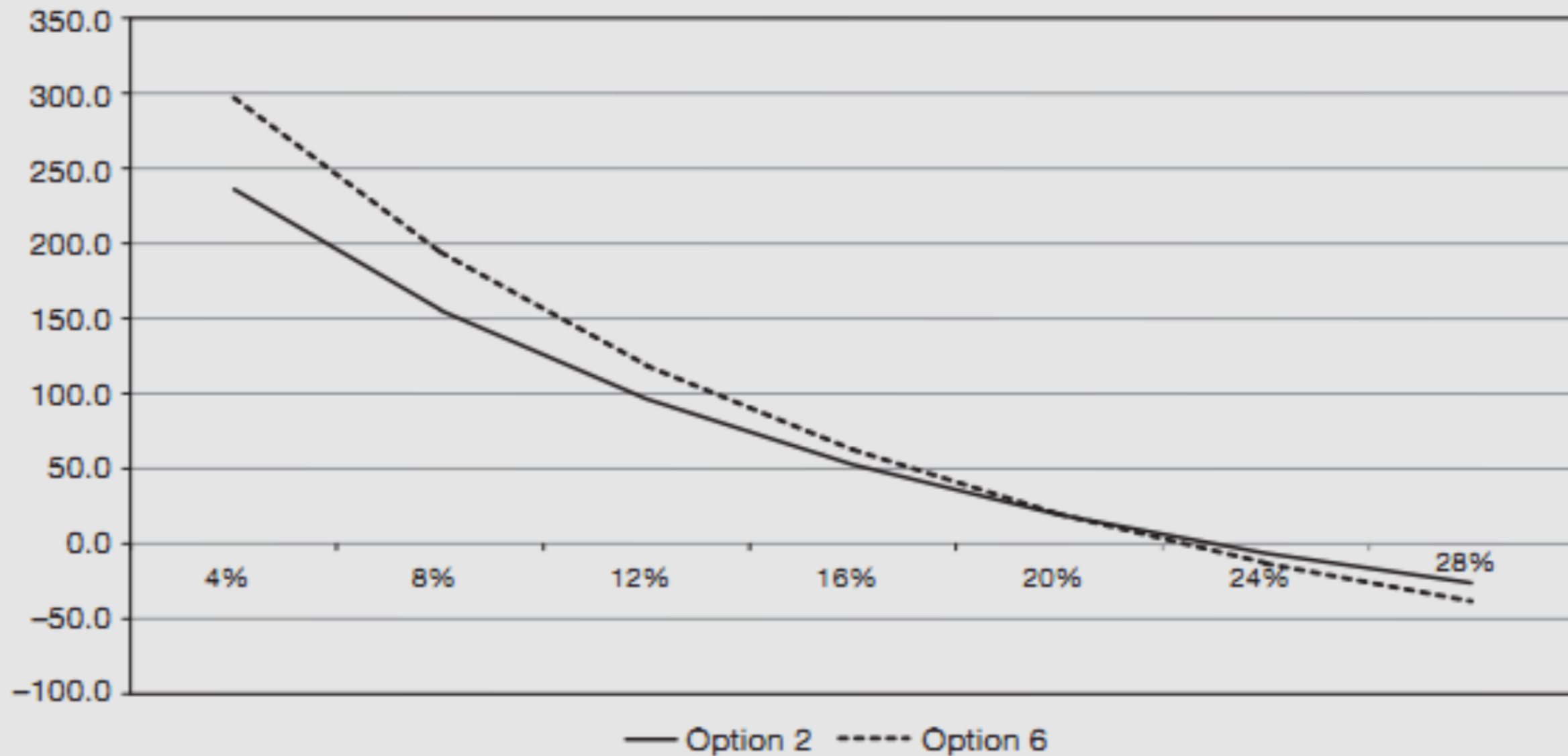


	A	B	C	D	E	F	G	H	I	J	K	L
1	OPTION 1: BANGKOK - NO CONCESSIONS											
2												
3	Project Cash Flow											
4												
5		1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
6	Investment Costs											
7	Fixed Investment	-282.5										
8	Working Capital		-69.3									69.3
9	TOTAL	-282.5	-69.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	69.3
10												
11	Operating Costs											
12	Raw Materials		-117.6	-235.2	-235.2	-235.2	-235.2	-235.2	-235.2	-235.2	-235.2	-235.2
13	Labour		-43.5	-43.5	-43.5	-43.5	-43.5	-43.5	-43.5	-43.5	-43.5	-43.5
14	Fuel, Water		-7.2	-14.4	-14.4	-14.4	-14.4	-14.4	-14.4	-14.4	-14.4	-14.4
15	Spare Parts		-22.4	-22.4	-22.4	-22.4	-22.4	-22.4	-22.4	-22.4	-22.4	-22.4
16	Insurance and Rent		-6.0	-6.0	-6.0	-6.0	-6.0	-6.0	-6.0	-6.0	-6.0	-6.0
17	TOTAL		-196.7	-321.5	-321.5	-321.5	-321.5	-321.5	-321.5	-321.5	-321.5	-321.5
18												
19	Revenues											
20	Cost Savings		139.6	279.2	279.2	279.2	279.2	279.2	279.2	279.2	279.2	279.2
21	Yarn Sales		69.8	139.6	139.6	139.6	139.6	139.6	139.6	139.6	139.6	139.6
22	TOTAL		209.4	418.8	418.8	418.8	418.8	418.8	418.8	418.8	418.8	418.8
23												
24	NET CASH FLOW	-282.5	-56.6	97.3	97.3	97.3	97.3	97.3	97.3	97.3	97.3	166.6
25												
26	Discount Rates	4%	8%	12%	16%	20%						
27	NPV	405.3	259.8	152.0	70.7	8.3						
28												
29	IRR	21%										

	A	B	C	D	E	F	G	H	I	J	K	L
33	Private Cash Flow											
34												
35			1	2	3	4	5	6	7	8	9	10
36	FINANCING	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
37	Principal											
38	Loan	160.1	-27.3	-29.5	-31.8	-34.4	-37.1					
39	Overdraft		60.0									-60.0
40	Interest											
41	Loan		-12.8	-10.6	-8.3	-5.7	-3.0					
42	Overdraft			-6.0	-6.0	-6.0	-6.0	-6.0	-6.0	-6.0	-6.0	-6.0
43	NET FINANCING FLOW	160.1	19.9	-46.1	-46.1	-46.1	-46.1	-6.0	-6.0	-6.0	-6.0	-66.0
44												
45	NCF (Equity, Pre-Tax)	-122.4	-36.7	51.2	51.2	51.2	51.2	91.3	91.3	91.3	91.3	100.6
46												
47												
48	TAXES											
49	Revenues		209.4	418.8	418.8	418.8	418.8	418.8	418.8	418.8	418.8	418.8
50	Operating Costs		-196.7	-321.5	-321.5	-321.5	-321.5	-321.5	-321.5	-321.5	-321.5	-321.5
51	Depreciation		-26.4	-26.4	-26.4	-26.4	-26.4	-26.4	-26.4	-26.4	-26.4	-26.4
52	Interest on Loans		-12.8	-16.6	-14.3	-11.7	-9.0	-6.0	-6.0	-6.0	-6.0	-6.0
53	PROFITS BEFORE TAX		-26.6	54.2	56.6	59.1	61.9	64.8	64.8	64.8	64.8	64.8
54	TAX CASH FLOWS		13.3	-27.1	-28.3	-29.6	-30.9	-32.4	-32.4	-32.4	-32.4	-32.4
55												
56												
57	EQUITY AFTER TAX											
58	PRIVATE CASH FLOW	-122.4	-23.4	24.1	22.9	21.6	20.3	58.9	58.9	58.9	58.9	68.2
59												
60	Discount Rates	4%	8%	12%	16%	20%						
61	NPV	154.5	88.7	40.8	5.4	-21.1						
62												
63	IRR	17%										

Discount Rates (for NPV): Market analysis		4%	8%	12%	16%	20%	RANK	IRR	RANK
BANGKOK	Option 1	405.3	259.8	152.0	70.7	8.3	3	21%	3
	Option 2	568.5	393.4	263.3	164.8	89.0	1	26%	1
	Option 3	405.3	259.8	152.0	70.7	8.3	3	21%	3
SOUTHERN THAILAND	Option 4	341.4	200.9	96.9	18.5	−41.7	4	17%	4
	Option 5	504.5	334.5	208.2	112.6	39.1	2	23%	2
	Option 6	341.4	200.9	96.9	18.5	−41.7	4	17%	4
	Option 7	504.5	334.5	208.2	112.6	39.1	2	23%	2

Discount Rates (for NPV): Private analysis		4%	8%	12%	16%	20%	RANK	IRR	RANK
BANGKOK	Option 1	154.5	88.7	40.8	5.4	−21.1	6	17%	6
	Option 2	236.1	155.4	96.4	52.5	19.2	4	23%	3
	Option 3	361.2	252.9	173.3	113.8	68.6	2	29%	2
SOUTHERN THAILAND	Option 4	119.1	53.7	6.2	−28.9	−55.2	7	13%	7
	Option 5	200.7	120.5	61.8	18.2	−14.8	5	18%	5
	Option 6	296.7	194.0	118.6	62.4	19.8	3	22%	4
	Option 7	459.9	327.5	229.9	156.5	100.5	1	31%	1



Switching Options 2 and 6: NPV Schedules

Alternative investment criteria

Q&A section

